

Csound Brain Lab: A Web Environment for Performing and Researching Brain-Computer Music Interfaces

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Abstract. The Csound Brain Lab (CBL) is a web-based application that lets anyone get started performing and researching Brain-Computer Music Interfaces (BCMI). It combines the flexible audio-synthesis building blocks of browser-based Csound with one-click connectivity to Muse and OpenBCI headsets through a local bridge, behind a beginner-friendly interface that lets users visualize and listen to their own live brain data, or experiment with a built-in EEG simulator when no sensor is at hand. The system also embeds both remote and on-device large-language models that perform multimodal analysis of EEG and musical material, helping novices find meaningful correlations between brain activity and sound. We describe the mapping layer, in which a chosen *target state*, a continuously measured *affinity* to it, and musical parameters expressed as interpolated endpoint pairs together decide what an instrument plays; and we report four preliminary research studies and two EEG composition and performance systems now being prepared with it, each packaged as an application of its own.

Keywords: Csound, BCMI, EEG, Brain-Computer Music Interface, sonification, deep listening, music education

1 Introduction

Brain-Computer Music Interfaces have long promised to turn brain activity into music, but the path from curiosity to a working setup has been steep: acquire a headset, connect it, route the data into a music program, design a mapping from scratch, capture brain and audio, and only then analyze any correlation. The Csound Brain Lab (CBL) collapses that pipeline into a single webpage: from a cold browser tab, three clicks (connect a headset or the simulator, choose an instrument, start listening) reach a playing instrument. No plugin, no DAW and no Csound install: the synthesis engine (Csound 7 through `@csound/browser` [1]), the visualizers, the recording layer and one of the two AI models all run inside the browser tab, and only the Bluetooth link needs a process of its own, so the apparatus that once required a lab now runs beside a browser tab.

The application has three destinations: *EEG*, the signal itself; *Create*, a single library of everything a user makes or is given; and *Settings*, where a profile decides how much of the system a user is handed.

2 The EEG Source

CBL pairs a small local server with a browser front end. Bluetooth is owned by the server, so the browser never negotiates with a radio. A native Swift binary built on LibMuse streams Muse 2 and Muse S [4] (TP9, AF7, AF8, TP10 at 256 Hz, referenced to FPz); BrainFlow [2] opens an OpenBCI Ganglion [5]; Mind Monitor arrives as OSC [3]. The bridge relays the signal to the browser over a WebSocket at 30 Hz, each message carrying every full-rate sample since the last one and the device's sampling rate, and can re-emit as OSC. A simulator synthesizes five band-power profiles (relaxed eyes-closed, meditative, focused, drowsy, aroused) so classrooms can work without hardware.

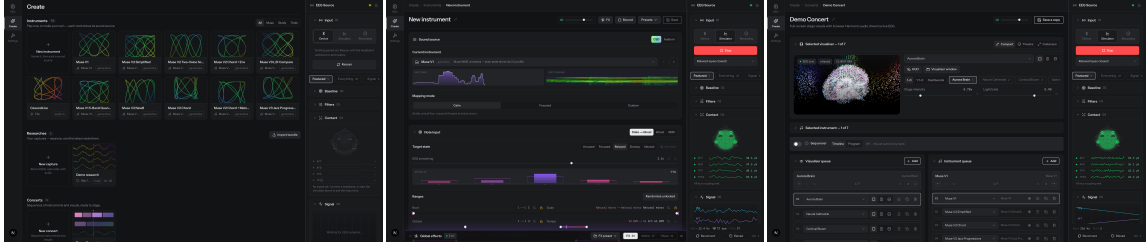


Fig. 1. Left: the library. Center: an instrument with the simulator running: the Csound orchestra with the spectrum and spectrogram of its own output, and the note-input container, where a target state, the affinity to it and the interpolated ranges decide what is played. Right: a concert, the Aurora Brain visualizer on the stage and the two independent queues below it. The EEG source column is docked at the right of every screen.

The EEG source column runs down the right of every screen in the order the signal travels. Scalp EEG is easily corrupted by mains hum, motion, muscle, eye-blinks and poor contact, so it is conditioned at the device rate by seven toggleable stages (detrend, re-reference, bandpass, notch, smoothing, artifact gate, log transform) under five whole-chain presets.

Conditioning. After a 3.9 s leaky-integrator detrend, channels are re-referenced, by default to linked mastoids (the mean of TP9 and TP10), which keep the frontal pair symmetric for asymmetry work [18,19,25]; a common-average option is offered for denser montages [17]. The bandpass is 1–45 Hz (two cascaded second-order Butterworth sections) and the notch a $Q = 30$ bell at 50 or 60 Hz, both causal, as a live instrument requires [15,16]: the resulting delay, about 230 ms at 1 Hz and 7 ms above 10 Hz, is stated rather than hidden.

Artifacts. The gate reads every sample before the reference and flags a channel exceeding $100 \mu\text{V}$ or moving faster than $10 \mu\text{V}/\text{ms}$, thresholds drawn from published Muse cleaning [20,21]. It never alters a sample: the flag travels downstream with every recorded sample and band-power packet, as EEG/MEG reporting guidelines recommend [29,33,34], rather than being silently repaired [30].

Band power. Band powers refresh ten times a second: the newest 4 s are cut into 2 s Hann segments at 50% overlap and averaged into a one-sided spectral density, following the window used in Muse validation work [19,22], then integrated over five standard edges (δ 1–4, θ 4–8, α 8–13, β 13–30, γ 30–45 Hz). Relative power is each band’s share of the 1–45 Hz total, absolute power is dB relative to $1 \mu\text{V}^2$ (gamma on dry electrodes is largely muscle [32]), and every packet names its method and units [34]. Derived features use linear power: frontal alpha asymmetry [25], engagement as $\beta/(\alpha + \theta)$, and theta/beta as a log-ratio that tracks the aperiodic slope [23,28].

Baselines. Normalization is a named baseline library rather than a filter stage. A timed capture (60 s by default) samples the band-power stream after a 5 s settling period and refuses to seal a profile from a stalled or short feed. Each profile stores the mean and standard deviation of every band’s centred log-ratio, since the five shares sum to one and log power is what resting-alpha analysis needs [26,27], along with the eyes state and conditioning in force. The active baseline z-scores live band powers in that space without mutating them, and is shared between the EEG surfaces and research captures, so that “high alpha” means the same thing across people, sessions and machines, as neurofeedback requires [33].

3 Instruments and How the Brain Plays Them

Every instrument answers one question, what is making the sound, in one of two ways: a Csound orchestra or audio the user brings. The orchestra branch is a single searchable list of 51 entries: 28 Cabbage [6] study orchestras loaded as genuine .csd files with their widgets parsed into on-screen controls; the Muse NIME orchestras and the Harmonic workstation, with its six variants, use `chnget` in place of `OSC` and `massign`. All receive the conditioned signal the same way, as twenty control channels named `nv_<band>_<channel>` beside raw, motion, PPG, fNIRS and CC channels. The audio

branch is a file, one of ten public-domain recordings or a microphone; every source plays through a shared rack of 34 global effects.

CBL’s central abstraction is a five-state taxonomy (aroused, focused, relaxed, drowsy, neutral), each state a profile of band shares summing to one. The performer picks a *target state*; the system reports *affinity*, a number in $[0, 1]$ scoring the live relative spectrum against that profile, drawn as five band columns with the target’s share as an outline behind the player’s (Fig. 1). Every brain-driven parameter is a pair of endpoints on one slider: a non-match value at zero affinity, a match value at one, a needle between them, a lock and a manual override, so that reaching the state is audible as the music travels from one endpoint to the other. Notes come from the state through a shared phrase generator, the Ghost (its parameters change only at phrase boundaries); from the Ghost free-running; or from a MIDI keyboard. Two authored modes, Calm and Focused, ship for all 51 orchestras as mappings of the rate of motivic development; Custom hands the endpoints to the user. The effects rack’s *Matrix* also routes 23 EEG and motion sources straight to Csound channels with per-channel invert, gain and smoothing, and *samphold* inside the orchestras turns a wandering value into a decision the music can rest on. The performer conducts; the brain sings.

The endpoint pair is CBL’s answer to a choice every brain-driven instrument makes: whether a signal drives sound directly or through a constrained, contextualized layer. A direct mapping wires a band power straight to pitch, loudness or brightness, so a rising number is simply heard rising, and leaves the performer’s ear responsible for musicality when the signal spikes on a blink or wanders. The endpoint pair instead fixes both outcomes in advance, so the result always falls inside a range the instrument’s composer has already voiced as music: a real transfer of agency, since the performer is now conducting the EEG rather than playing it, while key, register and timbre, the decisions that used to require a musician’s hands, now live in the two endpoints. Scoring affinity against an individual, z-scored baseline rather than a raw threshold extends the same idea across people: two performers reaching for the same target state hear analogous music even though their raw band powers differ.

4 Concerts

A concert sequences what is heard and what is seen. CBL offers 20 visualizer engines in two families: ten EEG-only (e.g. Aurora Brain, a luminous head-space silhouette whose lobes ripple per band) and ten audio-reactive (e.g. Pulse Rings, halos driven by audio RMS and the γ/β ratio), any of which can share the stage with a research dashboard, split, full-bleed, or sent to a chrome-free second window for a projector. The two tracks, visualizers and instruments, run as independent queues, each moved by arrow keys, on-screen buttons or a mapped MIDI controller. An optional sequencer advances a track on a timeline or on the signal itself, firing on a band change, a threshold crossing, or a state change, so a set can be driven by the clock, the performer’s hand, or the performer’s mind.

5 Research: Captures, Dashboards, and Analysis

A capture is the central unit of the research surface: a customizable dashboard, an optional audio source, synchronized EEG-and-audio takes, and an AI analysis. The dashboard assembles 19 display types (raw EEG, spectrogram, FFT, 3D waterfall, band history, brain state, PPG, fNIRS and more) in one- to three-column layouts, and the same 19 feed the EEG source column: where the performance visualizers are meant to be felt, these are meant to be read.

Within a capture each *take* is a sealed recording, chosen from a row of chips and shown first as its band powers over time, with markers and notes; audio recorded, attached or bounced from a Csound orchestra is drawn on the same time axis. Takes export as CSV or as a self-contained bundle another machine can import, and a demonstration capture with simulated EEG lets the whole workflow be tried before any hardware is paired.

The cloud analysis feeds the synchronized audio and EEG to Gemini 3.5 Flash [7], which returns three coordinated per-window analyses, musical, EEG, and correlation, arriving with a browser-computed spectrogram and time-aligned band-power plot, exportable as a PDF matching the on-screen report. An OSC tool emits the EEG to Max, Pure Data, TouchDesigner or Csound. Finally, *Live AI* runs a local Gemma 4 E2B model [8,9] in the browser through WebGPU: every two seconds it labels the recent band powers with a short tag and rationale, on a timeline aligned with the live band chart, with no API calls and nothing leaving the machine.

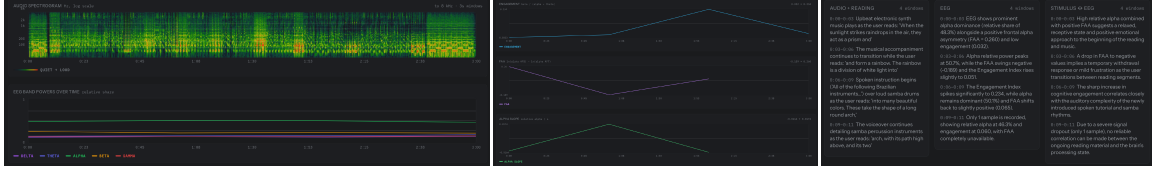


Fig. 2. From a saved report. Left: the audio the take carried, as a log-frequency spectrogram, over the EEG band powers on the same clock. Center: the derived features sent with them: an engagement index, frontal alpha asymmetry, and the slope of relative alpha. Right: the multimodal analysis in the model’s own words, describing, for each three-second window, what was heard and read, what the EEG did, and what the two have to do with each other.

6 Putting the Human Into the Experience

For all its filters and models, CBL’s deepest purpose is to put the human back at the center of computer music, in the tradition of two great listeners: Max Mathews [13], who sought instruments that liberate the musical mind, and Pauline Oliveros, whose *Deep Listening* [12] treats attention itself as transformative. CBL rewards and contextualizes attention in a feedback process, the music noticeably responding to how one mentally engages with it. Consider Shiyong, a young musician who, through the system, can listen to the brains of the people she loves, whether a mother, a six-year-old brother, or an elder who after a stroke can no longer reach others through words: a way of giving voice to the voiceless. The scientific and aesthetic visualizers let her make sense of what she hears, and the musical controls let her turn that understanding into expression. Beyond this therapeutic and rehabilitative research, Shiyong now composes and performs brainwave solos and duets. Another concert use is by a young, award-winning pianist, Yuwen, who uses her brainwaves to trigger, play, and modify a collection of Csound effects that transform the sound of the acoustic piano as she plays. Both performers have original CBL-supported compositions premiering at the International Csound Conference in Trapani, Sicily.

That promise is being tested by other hands. Four preliminary studies are in preparation, and each runs as its own application, built from a profile checked into the repository that decides which modes, instruments and research tools it holds, so that a researcher is handed a system that supports the functionality needed to answer their specific research question, rather than a laboratory. In Wentao’s Poetry and Music study, the subjects read from a paced presenter that carries a passage as a moving highlight or one word at a time, held at a fixed anchor to keep the eyes still; the reader and recorder start on the same edge, so the page and the electrophysiology share an origin. In Xudong’s sleep study, subjects listen to noise of various colors, nature sounds, favorite music, generative Csound soundscapes, or Csound compositions composed in response to their EEG as they enter different sleep states. In Luoxuan’s study, the research hub is paired with fourteen EEG-driven orchestras; the focus is an explicit target profile the music is scored against. In Jiahe’s study, the focus is musical memory: starting from the listener’s own favorite and performance recordings, the subject reflects on the music of their life alongside their own EEG and self-reported memories. What the system gives each of them is a unique, customized research apparatus. And it points toward a music education built on audio artistry, on understanding what sound *can be*, placing a student at the center of an instrument whose voice is, quite literally, their own mind.

7 Next Steps

We are lowering the barrier to *making* instruments. Loading one’s own .csd is already possible; next are a browseable Csound Web DSP catalog and an LLM that adapts an orchestra to receive EEG input from a plain-language description. An EEG-as-smart-controller mode will let the on-device model’s labels and derived features (engagement, fatigue, dominant band, transitions) steer higher-level behavior rather than raw parameters, in the spirit of systems such as NeuroSkill, which model a user’s state of mind in real time to drive a proactive agent [10,11]. We are also preparing code and tutorials for release, and exploring multi-brain analysis relating the EEG of a listener and a performer: because the stream can be shared, CBL opens the door to *brain-net* collaboration, many minds shaping one evolving voice.

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